

Software Requirements Specification (SRS) - School Management System

1. The Purpose of this Document

This Software Requirements Specification (SRS) defines the requirements for the School Management System (SMS). It documents the functional, non-functional, and design constraints agreed upon by the development team and the stakeholders. The system aims to centralize the daily operations of a school, including user management, attendance, grading, communication, and scheduling, replacing manual or disparate old systems.

2. The Stakeholders

The following table identifies and summarizes the key stakeholders and their primary areas of interest in the system:

Stakeholder Category	Identified Stakeholder(s)	Primary Interest in the System
Internal	School Administrator, Admins, Teachers	Data management, class schedules, announcements, attendance, grades, study materials
External	Parents, Software Developers	Child's grades/attendance, system feature upgrades, security
Operational	Students, Teachers, Parents	Checking grades/attendance/announcements, entering grades, material sharing, tracking progress

Stakeholder Category	Identified Stakeholder(s)	Primary Interest in the System
Executive	School Principal, Vice Principal, Head of Departments	Tracking teacher efforts, decision-making based on system data, reviewing academic progress

3. Mandates, Constraints, and Non-Functional Requirements

3.1. Design Constraints and Limitations

The system must adhere to the following limitations and technical constraints:

- **Accessibility:** The system shall be a web-based application, accessible through a standard web browser on multiple devices (desktop, laptop, tablet).
- **Connectivity:** The system shall require an active internet connection to function.
- **Out of Scope:** The system shall not include school fees or payment processing functionality.
- **Platform:** There will be no dedicated native mobile application in the initial release.

3.2. Look and Feel Requirements (Usability/Aesthetics)

The design of the application must prioritize user experience (UX):

- The interface design should be visually appealing, clear, and modern (as per Questionnaire results - Looks priority).
- The system should provide simple labels and help text to guide the user (as per Questionnaire results - Guidance priority).

- The system shall ensure page loads quickly to prevent user frustration.

3.3. Performance Requirements (Speed and Accuracy)

- Transaction Speed: The latest grade shall appear in the system immediately after a teacher finishes marking it.
- Report Generation: The system shall generate attendance and grade reports easily and quickly for administrative review.
- Calculations: The system shall automatically calculate and display overall grades/averages, accommodating customizable weighting.

3.4. Security and Auditing Requirements

- Authentication: The system shall require users (Admin, Teacher, Student, Parent) to log in with a unique username and password.
- Authorization: The system shall enforce role-based access control:
 - Admin: Full access to manage all modules.
 - Teacher: Access to their classes, attendance, and grades modules.
 - Student: Access to personal grades, schedule, and announcements.
 - Parent: Access to their child's grades, attendance, and relevant announcements.
- Data Protection: The system shall store all student data securely in a central database.
- Auditing: The system shall log on to user activities (e.g., login, data updates).
- Recovery: The system shall automatically back up data periodically.

4. Functional Requirements (Use Cases & Features)

Functional requirements detail the processes the system must perform. They are structured by the identified Use Cases (UCs).

4.1. UC: Manage User Accounts

Actor: Admin

ID	Functional Requirement
FR-UAM-100	The system shall allow the Admin to add, view, edit, and delete user accounts (Admin, Teacher, Student).
FR-UAM-110	The system shall allow the Admin to assign teachers to specific classes/subjects.
FR-UAM-120	The system shall allow the Admin to set different access rights based on user roles.
FR-UAM-130	The system should provide a password reset function for users.

4.2. UC: Attendance Management

Actor: Teacher (Record), Parent (View), Student (View), Admin (View/Report)

ID	Functional Requirement
FR-ATT-100	The system shall allow Teachers to record attendance (Mark students present/absent) for each class session.
FR-ATT-110	The system shall allow Parents to view their child's attendance records.
FR-ATT-120	The system shall allow Students to view their own attendance history.

ID	Functional Requirement
FR-ATT-130	The system shall allow the Admin to view attendance records for all classes.

4.3. UC: Grades Management

Actor: Teacher (Enter/Update), Student (View), Parent (View), Admin (Report)

ID	Functional Requirement
FR-GRD-100	The system shall allow Teachers to enter and update student grades for each subject.
FR-GRD-110	The system shall calculate and display overall grades/averages automatically.
FR-GRD-120	The system shall allow Students to view their grades securely.
FR-GRD-130	The system shall allow Parents to view their child's grades ⁵³ .
FR-GRD-140	The system shall allow Admins to generate grade reports for review ⁵⁴ .

4.4. UC: Announcements and Communication

Actor: Admin (Create/Send), Teacher (Create/Post), Student (View/Receive), Parent (View/Receive) 5555555555555555

ID	Functional Requirement
FR-COM-100	The system shall allow the Admin to create and send announcements to all users or selected groups (e.g., teachers, students, parents).
FR-COM-110	The system shall allow Teachers to post class-related notices viewable by their students.
FR-COM-120	The system shall display announcements on the dashboard of each user.
FR-COM-130	The system should offer multiple notification methods for urgent announcements (in-system alert, email, SMS/Text).
FR-COM-140	The system shall provide built-in direct messaging or a shared class announcement board for teacher-student communication.

4.5. UC: Class Scheduling

Actor: Admin

ID	Functional Requirement
FR-SCH-100	The system shall allow the Admin to gather scheduling inputs (e.g., available classes, rooms, teachers).

ID	Functional Requirement
FR-SCH-110	The system shall allow the Admin to select student group, subject, assign a teacher, time slot, and room ⁶³⁶³⁶³⁶³ .
FR-SCH-120	The system shall check for conflicts (Teacher/Room/Group unavailability) before saving a new session to the draft schedule.
FR-SCH-130	If a conflict is found, the system shall allow the Admin to re-assign/edit the session details.
FR-SCH-140	The system shall allow the Admin to finalize and publish the Master Schedule once complete.
FR-SCH-150	The system shall notify teachers of the new schedule upon publishing ⁶⁷ .
FR-SCH-160	The system shall make the schedule viewable to students upon publishing ⁶⁸ .

4.6. Other Core Functions

ID	Functional Requirement
FR-DMR-100	The system shall allow Admins to generate simple reports on attendance, grades, and student performance ⁶⁹⁶⁹⁶⁹⁶⁹ .
FR-DMR-110	The system shall provide search and filter options to quickly find records (e.g., filter by date, subject, user role) ⁷⁰⁷⁰⁷⁰⁷⁰ .
FR-MAT-100	The system shall allow Teachers to upload study materials for students to view or download ⁷¹⁷¹⁷¹⁷¹ .
FR-MAT-110	The system shall allow Students to access and download all learning materials in one place ⁷²⁷² .

5. Data Structure and Attributes

This section defines the key data entities and the core attributes they must store, based on the requirements for management, attendance, and grading.

5.1. User Entity (Shared)

Attribute	Description	Data Type	Notes
UserID	Unique identifier for each user.	Integer	Primary Key, Auto-increment
Username	Unique name for login.	String	Must be verified before access (FR-UAM-100) ⁷⁴ .

Attribute	Description	Data Type	Notes
PasswordHash	Secured password storage.	String	
Role	User role (Admin, Teacher, Student, Parent).	Enum	Controls access rights (FR-UAM-120) ⁷⁵ .

5.2. Student Entity

Attribute	Description	Data Type	Notes
StudentID	Unique ID for the student.	String/Int	Used for viewing grades/attendance (FR-UAM-100) ⁷⁶ .
ParentContactID	Link to the Parent's account.	Integer	Foreign Key to User (for Parent) ⁷⁷ .
ClassEnrollment	List of classes/subjects for the student.	List/Array	

5.3. Attendance Record Entity

Attribute	Description	Data Type	Notes
AttendanceID	Unique ID for the record.	Integer	Primary Key.
StudentID	Link to the Student.	Integer	Foreign Key.
ClassSessionID	Link to the scheduled session.	Integer	Foreign Key.
Date	Date of the class session.	Date	
Status	Present, Absent, Late.	Enum	Recorded by Teacher (FR-ATT-100) ⁷⁸⁷⁸⁷⁸⁷⁸ .

5.4. Grade Record Entity

Attribute	Description	Data Type	Notes
GradeID	Unique ID for the grade entry.	Integer	Primary Key.
StudentID	Link to the Student.	Integer	Foreign Key.
Subject	Subject name or ID.	String/Int	

Attribute	Description	Data Type	Notes
AssignmentName	Name of the assignment/test.	String	
Score	Raw score achieved.	Float	
Weighting	Weighting factor for calculations (if used).	Float	Used for automated averages (FR-GRD-110) ⁷⁹ .

6. Use Case Diagram and Activity Diagrams

To understand the scope and flow of the system, two key UML diagrams are referenced:

6.1. Use Case Diagram for School Management System

The diagram visually represents the relationships between the main Actors and the key Use Cases they interact with ⁸⁰⁸⁰⁸⁰⁸⁰.

Actor	Key Use Cases
Admin	Manage User Accounts ⁸¹⁸¹ , Add/Edit Student Info, Assign Teachers to Classes, Publish Schedule ⁸² , Generate Reports ⁸³⁸³ , Announcements and Communication ⁸⁴⁸⁴ .

Actor	Key Use Cases
Teacher	Record Attendance ⁸⁵⁸⁵ , Enter or Update Grades, Upload Study Materials ⁸⁶ , Announcements and Communication, Publish Schedule.
Student	View Grades ⁸⁷⁸⁷ , View Attendance ⁸⁸⁸⁸ , Announcements and Communication.
Parent	View Grades, View Attendance, Announcements and Communication ⁸⁹⁸⁹ .

6.2. Activity Diagram: Announcements and Communication

This diagram illustrates the step-by-step process flow for creating, sending, and viewing an announcement, including validation, multi-channel notification, logging, and optional reply functionality :

- Creator (Admin/Teacher): Logs in \$to\$ Opens New Announcement \$to\$ Selects recipients \$to\$ Writes message/Attaches files \$to\$ If Urgent? \$to\$ System validates ⁹⁰\$to\$ Saves announcement \$to\$ Posts \$to\$ System notifies via Dashboard/Email/SMS \$to\$ Receives "Sent Successfully" confirmation⁹¹.
- Recipient (Student/Parent): Views announcement \$to\$ Can Reply? (Decision) \$to\$ If Allowed \$to\$ Writes and sends reply \$to\$ Creator is notified of the reply.

6.3. Activity Diagram: Class Scheduling Process

This diagram outlines the process an Admin follows to create and publish the master class schedule :

- School Admin: Gathers Inputs \$to\$ Selects Group/Subject \$to\$ Assigns Teacher/Slot/Room \$to\$ System checks for Conflicts ⁹² \$to\$ If Conflict: Re-

assigns \$\\to\$ If No Conflict: Saves to Draft Schedule \$\\to\$ Is Whole
Schedule Done? \$\\to\$ If No: Loop back to Select Group \$\\to\$ If Yes:
Finalizes and Publishes \$\\to\$ Notifies Teachers/Makes viewable to Students
⁹³ \$\\to\$ End.

7. Volere Shell Template

Requirement Type	Description	Rationale	Fit Criterion (Measurable Test)
Functional (FR-ATT-110)	The system shall allow Parents to view their child's attendance records ⁹⁴⁹⁴⁹⁴⁹⁴ .	This ensures parents are aware of their child's school presence, fulfilling a core stakeholder need ⁹⁵⁹⁵⁹⁵⁹⁵ .	Upon login, Parent must be able to view attendance reports for their linked child(ren) for the last 30 school days with a successful response time under 3 seconds ⁹⁶ .
Functional (FR-GRD-100)	The system shall allow Teachers to enter and update student grades for each subject ⁹⁷⁹⁷⁹⁷⁹⁷ .	This replaces the slow, error-prone manual grade entry, saving teachers time and keeping data up-to-date ⁹⁸ .	A Teacher can successfully enter 20 grades for 20 unique students in a single class within 90 seconds, and the data must

Requirement Type	Description	Rationale	Fit Criterion (Measurable Test)
			persist immediately ⁹⁹ .
Non-Functional (Usability)	The latest grade shall appear in the system immediately after a teacher says they have finished marking it ¹⁰⁰ .	Students and parents need immediate feedback to track progress and reduce confusion ¹⁰¹¹⁰¹¹⁰¹ .	The average latency from teacher clicking 'Save' on a grade record to the grade appearing on the Student/Parent View screen must be less than 1 second ¹⁰² .
Non-Functional (Security)	The system shall log on to user activities (e.g. login, data updates) ¹⁰³ .	This is critical for auditing, system debugging, and meeting compliance expectations of school executives ¹⁰⁴ .	All 'Login', 'Update Grade', and 'Delete Account' actions must generate an uneditable audit log entry including UserID, Timestamp, ActionType, and DataAffected ¹⁰⁵ .

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8. Detailed Functional Requirements (Use Case Flows)

8.1. UC: Class Scheduling - Conflict Management Details

The core of the scheduling module lies in the system's ability to check for and handle resource conflicts during the "Assign Teacher Time Slot and Room" step¹¹¹¹¹.

ID	Description	Pre-conditions	Post-conditions
FR-SCH-200	Execute Conflict Check ² The System shall automatically check the availability of the selected Teacher, Room, and Student Group against the Draft Schedule when a new session assignment is made ³³ .	Admin has entered Student Group, Subject, Teacher, Time Slot, and Room ⁴⁴⁴ .	System returns No Conflict (FR-SCH-210) or Conflict Found (FR-SCH-220).
FR-SCH-210	Save Class Session to Draft Schedule ⁵ If no conflict is found (No Conflict path) ⁶ , the system shall save the new class session	Conflict Check (FR-SCH-200) result is No Conflict.	The class session is permanently recorded in the Draft Schedule.

ID	Description	Pre-conditions	Post-conditions
FR-SCH-240	Finalize and Publish Master Schedule ¹⁶Upon Admin confirmation ("Yes - Schedule Complete")¹⁷, the system shall promote the Draft Schedule to the official Master Schedule¹⁸.	Completion Check (FR-SCH-230) result is Yes.	The Master Schedule is published. System proceeds to notify users (FR-SCH-250, FR-SCH-260).
FR-SCH-250	Notify Teachers of New Schedule ¹⁹The system shall automatically send an internal notification or email to all Teachers confirming the publication of the new schedule²⁰.	FR-SCH-240 complete.	Teacher accounts are flagged to view the new schedule.
FR-SCH-260	Make Schedule Viewable to Students ²¹The system shall make the newly published Master Schedule accessible to Students (and Parents	FR-SCH-240 complete.	Students can view the updated schedule (FR-SCH-160).

ID	Description	Pre-conditions	Post-conditions
	via Student profile) ²²²²²²²² .		

8.2. UC: Announcements and Communication - Multi-Channel Delivery

The communication module enables Creators (Admin or Teacher) to send announcements using multiple methods, ensuring delivery tracking and adherence to notification preferences²³²³²³²³²³²³²³.

ID	Description	Trigger	Delivery Channel
FR-COM-200	Message Validation ²⁴²⁴²⁴ The System shall validate the message content and recipient list for format/completeness before processing for delivery ²⁵²⁵²⁵²⁵²⁵²⁵²⁵²⁵ .	Admin/Teacher initiates send (post-FR-COM-100).	N/A
FR-COM-210	Post to Dashboard ²⁶ The System shall post the announcement content to the in-system dashboard of all relevant recipients ²⁷ .	FR-COM-200 is Valid; In-system notification selected ²⁸ .	In-system Dashboard/Feed.
FR-COM-220	Send Email Notification ²⁹²⁹ The System shall send an email containing the announcement to	FR-COM-200 is Valid; Email notification	Email (Standard or Instant) ³¹³¹³¹³¹ .

ID	Description	Trigger	Delivery Channel
	recipients who prefer or are selected for email notification ³⁰³⁰³⁰³⁰ .	selected (e.g., non-urgent).	
FR-COM-230	Send SMS Notification ³² The System shall send an SMS text message (containing critical information or a link) to recipients selected for SMS notification ³³ .	FR-COM-200 is Valid; SMS notification selected (e.g., urgent/SMS) ³⁴ .	SMS/Text message.
FR-COM-240	Log Delivery and Status ³⁵ The System shall update the delivery status and log the delivery activity to the database regardless of the channel used (dashboard, email, or SMS) ³⁶³⁶³⁶³⁶ .	Post-delivery (FR-COM-210, 220, 230).	Records Module (FR-DMR-100).

8.3. UC: Announcements and Communication - Reply Flow

The communication flow includes optional two-way interaction for recipients, adhering to the "Reply Allowed?" setting defined by the creator³⁷.

ID	Description	Pre-conditions	Flow Path
FR-COM-300	Reply Option Display ³⁸The Recipient view shall only display the "Reply" option if the original sender (Creator) allowed replies for that announcement³⁹.	Recipient views announcement (FR-COM-210 or 220 or 230).	If Allowed \$\\to\$ FR-COM-310. If Not Allowed \$\\to\$ FR-COM-330.
FR-COM-310	Write and Send Reply ⁴⁰The Recipient shall be able to compose and send a reply message to the Creator⁴¹.	FR-COM-300 result is Allowed.	Reply is sent to the System.
FR-COM-320	Receive and Store Reply ⁴²The System shall receive the reply message, store it in the database, and link it to the original announcement thread⁴³.	FR-COM-310 is complete.	System proceeds to notify original sender (FR-COM-340).
FR-COM-330	End Communication Flow (No Reply) ⁴⁴If replies are not allowed, the system shall terminate the communication	FR-COM-300 result is Not Allowed.	Communication flow ends (Activity Final Node).

ID	Description	Pre-conditions	Flow Path
	workflow for that recipient ⁴⁵ .		
FR-COM-340	Notify Original Sender of Reply ⁴⁶ The System shall notify the original Creator (Admin/Teacher) about the new reply received ⁴⁷ .	FR-COM-320 is complete.	Creator receives and views the reply notification ⁴⁸ .

9. Non-Functional Requirement Details

9.1. Non-Functional: Grade Calculation Logic (Clarity)

The system shall support flexible grade calculation methods for Teachers, with the default being Pre-defined categories with fixed weights⁴⁹.

ID	Non-Functional Requirement	Priority	Fit Criterion (Measurable Test)
NFR-CALC-100	Flexible Weighting Support The system shall allow the Teacher to use <i>Pre-defined categories with fixed weights</i> or <i>Customizable weights</i> for the final average calculation ⁵⁰ .	High	The Teacher settings panel must allow selection between "Fixed Weights" (e.g., Homework 20%, Test 50%) and "Custom Weights" (per assignment).

ID	Non-Functional Requirement	Priority	Fit Criterion (Measurable Test)
NFR-CALC-110	Instant Final Average The system shall compute and display the final overall course average for students immediately upon the latest grade update ⁵¹⁵¹⁵¹⁵¹ .	Critical	Calculate average for a class of 30 students with 10 assignments in ≤ 500 ms upon saving a new score.
NFR-CALC-120	Missing Work Alert The system should use a simple visual indicator (e.g., a red flag/icon) next to the assignment name to inform users of missing or past due assignments ⁵² .	Medium	When an assignment's due date passes and no score is recorded, a red flag icon must appear next to the assignment name in the Student's "Grades" view.

9.2. Non-Functional: System Accessibility and Interface (Usability)

ID	Non-Functional Requirement	Priority	Fit Criterion (Measurable Test)
NFR-ACC-100	Cross-Device Support The web interface shall be fully functional and responsive when accessed via standard web browsers across Desktop/Laptop, Tablet, and Mobile Phone screens ⁵³⁵³⁵³⁵³ .	Critical	All critical functions (Login, Record Attendance, View Grades) must be fully usable on a screen width of ≥ 375 pixels.

ID	Non-Functional Requirement	Priority	Fit Criterion (Measurable Test)
NFR-ACC-110	Schedule Visualization The system should provide an Interactive Calendar View for checking class schedules and assignment due dates ⁵⁴ .	Medium	The default schedule view must be an interactive calendar that allows navigation by week/month and displays assignments overlaid on class times.
NFR-ACC-120	Primary Teacher Dashboard The default Teacher dashboard shall prioritize displaying <i>Quick links to attendance/grade entry pages</i> ⁵⁵ .	High	The teacher dashboard must load with two prominent links/widgets leading directly to the current day's attendance sheet and the grade entry interface for active classes.

9.3. Non-Functional: Data Integrity and Disaster Recovery

ID	Non-Functional Requirement	Priority	Fit Criterion (Measurable Test)
NFR-DR-100	Periodic Backup The system shall perform automatic data backup periodically ⁵⁶ .	Critical	Full system backup must be executed daily at 2:00 AM local time, and the last seven backups must be retrievable.

ID	Non-Functional Requirement	Priority	Fit Criterion (Measurable Test)
NFR-DR-110	Data Loss Prevention If the internet connection is temporarily lost while entering data, the system shall save the data locally (if possible) and synchronize it once reconnected ⁵⁷ .	High	A user entering 10 attendance records in an offline state must have those records successfully synchronized and appear in the Master Schedule database once the connection is restored.

10.1. UC: Student Management (Admin/Teacher)

This module ensures accurate and secure handling of student information¹.

ID	Description	Actor	Requirement Details
FR-SM-100	CRUD Operations for Student Info The system shall allow the Admin to add, view, edit, and delete student information ² (e.g., name, ID, class, contact details ³).	Admin	This covers the basic Create, Read, Update, and Delete operations ⁴ .
FR-SM-110	Student List Viewing The system shall allow Teachers to view student lists for their assigned classes ⁵ .	Teacher	Teachers should only see students relevant to their assigned classes.

ID	Description	Actor	Requirement Details
FR-SM-120	Secure Data Storage The system shall store all student data securely in a central database ⁶ .	System	Addresses data integrity and security non-functional requirements.
FR-SM-130	Parent-Child Linkage The system shall link a Student's record to their corresponding Parent account(s) to enable grade and attendance viewing.	Admin, System	Necessary for Parent access to information (FR-ATT-110, FR-GRD-130) ⁷⁷⁷⁷⁷⁷⁷⁷ .

10.2. UC: Teacher Management (Admin)

This module covers the administrative control over teacher accounts and their roles/assignments⁸.

ID	Description	Actor	Requirement Details
FR-TM-100	CRUD Operations for Teacher Profiles The system shall allow the Admin to add, view, edit, and delete teacher profiles ⁹ .	Admin	Manages teacher accounts alongside student and admin accounts (FR-UAM-100) ¹⁰ .
FR-TM-110	Class/Subject Assignment The system shall assign teachers to specific classes or subjects ¹¹ .	Admin	Essential prerequisite for Assign Teachers to Classes use case.

ID	Description	Actor	Requirement Details
FR-TM-120	Self-Service Profile Update The system shall allow teachers to update their profile information (excluding roles or permissions) ¹² .	Teacher	Provides necessary autonomy for minor profile changes.

11. UC: Reports and Data Management

This section covers requirements related to viewing, exporting, and managing the aggregate data generated by the system¹³.

ID	Description	Actor	Requirement Details
FR-RPT-100	Generate Simple Reports The system shall generate simple reports on attendance, grades, and student performance ¹⁴¹⁴¹⁴¹⁴ .	Admin	Supports the Admin goal to check school performance easily ¹⁵ .
FR-RPT-110	Attendance Reports The system shall generate attendance reports based on selected dates or classes ¹⁶ .	Admin	Enables high-level attendance review.
FR-RPT-120	Grade Reports The system shall allow admins to generate grade reports for review ¹⁷ .	Admin	Enables high-level academic performance review.

ID	Description	Actor	Requirement Details
FR-RPT-130	Search and Filter The system should provide search and filter options to quickly find records ¹⁸ (e.g., filter by date, subject, user role, basic keyword search) ¹⁹ .	All Users	Improves efficiency when retrieving specific data or announcements.
FR-RPT-140	Data Export (Offline) The system should allow Teachers to download and export final grade and attendance data for offline record keeping ²⁰ .	Teacher	Preferred export formats include CSV and Excel ²¹ .

12. UC: Study Materials Management

This module allows teachers to share learning resources with students²²²²²²²²²²²²²²²²²².

ID	Description	Actor	Requirement Details
FR-MAT-100	Upload Materials The system shall allow Teachers to upload study materials (files, links, documents) for their students ²³²³²³²³²³ .	Teacher	Teachers control the sharing of course materials ²⁴ .
FR-MAT-110	Material Access The system shall provide Students with access to all learning materials in one centralized place, enabling them	Student	Reduces student confusion and ensures easy

ID	Description	Actor	Requirement Details
	to download or view materials anytime ²⁵²⁵²⁵²⁵²⁵²⁵²⁵ .		access to resources ²⁶ .
FR-MAT-120	Material Organization The system should organize study materials by subject and class.	System	Ensures students can easily find the materials they need (FR-DMR-110).

13. Supporting Requirements (Non-Functional & Technical)

13.1. Non-Functional: System Management and Security

Building on previous requirements, this focuses on maintenance and protective measures.

ID	Non-Functional Requirement	Priority	Rationale
NFR-SEC-200	Restrict Unauthorized Access The system shall restrict unauthorized access to sensitive data ²⁸ , ensuring users can only view data authorized by their role (e.g., Parent sees only their child's data; Teacher sees only their class data).	Critical	Protects privacy and maintains data integrity under a multi-user system.

ID	Non-Functional Requirement	Priority	Rationale
NFR-SEC-210	User Activity Log Detail The system shall log detailed user activities ²⁹ , including login, data creation, data updates, and account removal.	High	Supports audit compliance (Executive Stakeholder need) and provides traceability for data changes.
NFR-SYS-100	Platform Requirement The system shall be accessible through a standard web browser ³⁰ and support multiple devices (desktop, laptop, tablet) ³¹ .	Critical	Confirms accessibility scope and portability limitations ³²³²³²³² .
NFR-SYS-110	Connectivity Requirement The system shall require an active internet connection to function ³³³³³³³³ .	Critical	Defines the boundary of operation and is a known system limitation ³⁴³⁴³⁴³⁴ .

This concludes the detailed documentation of the system's functional and non-functional requirements based on the provided data.

I will continue the SRS document by detailing the deployment context and the required appendices, focusing on the team structure, glossary, and open issues, thus completing the comprehensive document structure.

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14. Deployment and Deliverables

This section outlines how the system will be built, tested, and deployed, defining the project team and technical environment.

14.1. Development and Team Structure

The development of the School Management System is carried out by the following team¹:

Name	Student ID	Role
Marwan Khaled ²	23-101296 ³	[To be defined]
Youssef Idrees ⁴	23-101077 ⁵	[To be defined]
Hany Yasser ⁶	23-101107 ⁷	[To be defined]
Omar Elsayed ⁸	23-101091 ⁹	[To be defined]
Yassin Sameh ¹⁰	23-101192 ¹¹	[To be defined]

14.2. Operating and Technical Environment

- Operating System/Access: Accessible through a standard web browser ¹²on multiple devices (desktop, laptop, tablet)¹³.
- Source Code Repository: The project uses a GitHub repository for version control and collaborative development¹⁴.

GitHub Repository Link: <https://github.com/Yousuf-Idrees/School-Managment-System.git> ¹⁵

- Connectivity: An active internet connection is required to function¹⁶¹⁶¹⁶¹⁶.

14.3. Documentation and Deliverables

The following formal documentation artifacts must be delivered as part of the project:

- Software Requirements Specification (SRS): This document.
- User Stories and Use Cases: Detailed user needs, roles, and decomposition¹⁷¹⁷¹⁷¹⁷¹⁷¹⁷¹⁷¹⁷.
- UML Diagrams:
 - Use Case Diagram for the system¹⁸.
 - Activity Diagram for Announcements and Communication¹⁹¹⁹¹⁹¹⁹.
 - Activity Diagram for Class Scheduling Process²⁰²⁰²⁰²⁰.
- Stakeholder Analysis: Identification of all stakeholders and their interests²¹.
- Requirements Elicitation Report: Justification and detail on techniques used (Interviews, Questionnaires, Document Analysis, Observation, Prototyping)²²²²²²²².

15. Appendices

15.1. Glossary of Terms

Term	Definition
Admin	School Administrator; controls the whole system and manages accounts ²³ .
Creator	The user role (Admin or Teacher) initiating an Announcement.
Recipient	The target user (Student or Parent) of an Announcement or Communication.
SRS	Software Requirements Specification; this document defining the system's requirements.

Term	Definition
System	The core School Management Software application.
Teacher	User role responsible for adding grades, attendance, and class information ²⁴ .
UML	Unified Modeling Language, used for system diagrams.

15.2. Requirements Traceability Matrix (RTM) Example

Requirement ID	Source (FR, NFR, Feature)	Test Case ID	Status
FR-UAM-100	Admin manages user accounts ²⁵	TC-UAM-001	To Do
FR-ATT-100	Teacher records attendance ²⁶	TC-ATT-001	To Do
FR-COM-130	Urgent notification via SMS/Email 27272727	TC-COM-003	To Do
NFR-ACC-100	Accessible through a web browser 28	TC-ACC-001	To Do

15.3. Unresolved Issues and Future Scope (Limitations)

This section documents outstanding decisions and known limitations to manage stakeholder expectations (Volere's Issue List).

ID	Issue/Limitation	Decision Needed	Future Scope
IS-001	Mobile Access ²⁹	No mobile app for now; limited to web browser on small screens ³⁰ .	Mobile application development (future phase) ³¹ .
IS-002	Payment Systems ³²	Will not include school fees or payment systems ³³ .	Integration with a third-party payment gateway (future phase).
IS-003	Grade Weighting ³⁴	Confirm the exact mathematical formulas for <i>Pre-defined categories</i> and <i>Customizable weights</i> with the school.	None (requires final stakeholder sign-off).

16. Requirements Elicitation Techniques and Data Samples

This section details the methodologies used to gather and validate the requirements documented in this SRS, ensuring comprehensive coverage from all stakeholder perspectives¹¹¹.

16.1. Elicitation Techniques and Justifications

The team utilized a mix of five techniques to ensure complete and correct requirements from all users—admin, teachers, and students².

- Semi-Structured Interviews: Conducted with the school principal, teachers, and admin staff³.
 - Justification: Helps collect detailed information about current processes, such as scheduling rules and data management, and

identified problems faced⁴. It allows for follow-up questions for more detail⁵.

- Questionnaires: Given to teachers and students⁶.
 - Justification: Collects opinions quickly from many users⁷. It covers system access, interface preferences, and communication needs, supporting interview results with user data⁸⁸⁸⁸.
- Document Analysis: Reviewing documents like class schedules, attendance sheets, and grade records⁹.
 - Justification: Documents the school's current management methods and helps find data storage and organization structures, defining the rules for the database and scheduling logic¹⁰¹⁰¹⁰¹⁰.
- Observation: Watching teachers and admin staff perform daily tasks (taking attendance, entering grades)¹¹.
 - Justification: Helps validate the real workflow, revealing where users face difficulties, which steps take time, and which actions can be automated¹².
- Prototyping: Creation of simple screen mockups or wireframes for main pages (dashboard, forms)¹³.
 - Justification: Allows users to visualize the system and provide early feedback, especially crucial for complex modules like class scheduling and communication¹⁴¹⁴¹⁴¹⁴.

17. Raw Data Sample: Stakeholder Questionnaire Excerpt

This excerpt demonstrates the type of structured data collected, which directly informed non-functional requirements and design choices (e.g., NFR-ACC-100, NFR-CALC-100, FR-COM-130).

17.1. General and Access (All Users)

Question	Options and User Preferences (Excerpt)	Directly Influenced Requirement
If web-based, where would you primarily access it from? ¹⁵	Desktop/Laptop (at school or home) ¹⁶ , Tablet/iPad ¹⁷ , Mobile Phone Browser ¹⁸	NFR-ACC-100 (Cross-Device Support) ¹⁹
To ensure the system is "easy to use," which is the single highest priority? ²⁰	Speed (Fewest steps) ²¹ , Looks (Visually appealing) ²² , Guidance (Simple labels/help text) ²³	NFR-ACC-120 (Dashboard Priority), Sec 3.2 (Look and Feel) ²⁴

17.2. Teacher-Specific: Attendance and Grades

Question	Options and User Preferences (Excerpt)	Directly Influenced Requirement
When you record attendance, which input method would be most efficient? ²⁵	List-based check box ²⁶ , Quick search for absent students ²⁷ , Integration with a seating chart diagram ²⁸	FR-ATT-100 (Efficient Recording) ²⁹
How would you prefer to see the scores weighted for the final average calculation? ³⁰	Pre-defined categories with fixed weights ³¹ , Customizable weights per assignment ³² , Total points system ³³	NFR-CALC-100 (Flexible Weighting) ³⁴

Question	Options and User Preferences (Excerpt)	Directly Influenced Requirement
Which format would you most prefer for downloading/exporting data? ³⁵	CSV ³⁶ , Excel ³⁷ , PDF Document ³⁸	FR-RPT-140 (Data Export) ³⁹

17.3. Communication and Announcements

Question	Options and User Preferences (Excerpt)	Directly Influenced Requirement
When receiving urgent school announcements, do you prefer...? ⁴⁰	Instant Email notification ⁴¹ , In-system banner/pop-up alert ⁴² , SMS/Text message ⁴³	FR-COM-130 (Multi-channel Urgent Delivery) ⁴⁴
What is the most important feature the system should have to help students and teachers communicate better? ⁴⁵	Built-in direct messaging functionality ⁴⁶ , A shared announcement board for each class ⁴⁷	FR-COM-140 (Communication Tool) ⁴⁸

18. Acceptance Criteria (Given-When-Then)

This section provides the testable, scenario-based acceptance criteria for key user stories, confirming that the system meets the functional requirements from the user's perspective. These scenarios are derived directly from the initial user stories.

18.1. User Management (Admin)

User Story ID	User Role	Scenario: Acceptance Criteria (Given-When-Then)	Related FR/NFR
US-A-001	Admin	Given the Admin is logged in, When they navigate to the 'Manage User Accounts' page , Then they can successfully add a new Teacher account with a role assignment and the new user can log in with valid credentials.	FR-UAM-100, FR-UAM-120

18.2. Attendance and Grading (Teacher)

User Story ID	User Role	Scenario: Acceptance Criteria (Given-When-Then)	Related FR/NFR
US-T-001	Teacher	Given the Teacher is logged in and views their current class roster ⁵ , When they select a student and mark their status as "Absent" , Then the attendance record is instantly saved , and the Parent can immediately view the updated absence record for that student.	FR-ATT-100, NFR-DR-110, NFR-SEC-200
US-T-002	Teacher	Given the Teacher is entering grades for an assignment , When they input the score for a student and save the record , Then the system must calculate and display the student's new overall course	FR-GRD-100, FR-GRD-110,

User Story ID	User Role	Scenario: Acceptance Criteria (Given-When-Then)	Related FR/NFR
		average, and that average must be updated on the student dashboard within \$1\$ second.	NFR-CALC-110

18.3. Viewing Information (Student/Parent)

User Story ID	User Role	Scenario: Acceptance Criteria (Given-When-Then)	Related FR/NFR
US-S-001	Student	Given the Student is logged in, When they navigate to the 'Grades' module , Then they can see a clear view of their individual assignment scores and final course average organized by course.	FR-GRD-120
US-P-001	Parent	Given the Parent is logged in , When they access the system, Then they can only view the grades and attendance records for their linked child(ren), and all other sensitive data must be restricted.	FR-GRD-130, FR-ATT-110, NFR-SEC-200

19. Detailed Communication Flows: Delivery Confirmation and Logging

Expanding on the workflow defined in the Communication Activity Diagram , the technical requirements for auditability and confirmation are detailed.

19.1. Delivery Confirmation (System \$to\$ Creator)

ID	Description	Pre-conditions	Success Criteria
FR-COM-400	Delivery Status Update The System shall receive confirmation from third-party services (Email/SMS) that the notification was sent or delivered.	Notification channels (FR-COM-220, FR-COM-230) are successfully triggered.	System proceeds to Update delivery status or Log delivery.
FR-COM-410	Send Confirmation to Creator After the System updates all delivery logs (FR-COM-240), it shall send a "Send confirmation to creator" message/notification.	Delivery logging is complete.	Creator receives the "Announcement Sent Successfully" message.

19.2. Auditing and Record Keeping (System \$to\$ Records)

The Records partition in the Communication Activity Diagram is mandatory for audit trail purposes.

ID	Description	Triggering Event	Audited Information (Example)
FR-AUD-100	Log Announcement Activity The System shall log the creation and content of every	Save announcement (FR-COM-200)	Creator ID, Timestamp, Message Content, Recipients List, "Urgent" Flag.

ID	Description	Triggering Event	Audited Information (Example)
	announcement sent.		
FR-AUD-110	Log Delivery Activity The System shall log the delivery attempts for all recipients across all channels.	Update delivery status or Log delivery (FR-COM-240)	Recipient ID, Channel Used (Dashboard/Email/SMS), Delivery Status, Timestamp.
FR-AUD-120	Log Reply Activity The System shall log the content and metadata of any reply received.	Receive reply and store in database (FR-COM-320)	Reply Content, Reply Sender ID, Original Announcement ID, Timestamp.

20. System Architecture and Technology Stack

This section defines the high-level technical decisions regarding the components and deployment environment for the School Management System (SMS).

20.1. Technology Decisions

The system will utilize a modern web application architecture designed for responsiveness and scalability, directly addressing the requirement for accessibility across multiple devices.

- Front-end (Client-side): A responsive framework (e.g., React, Angular, or Vue.js) to ensure the web interface is fully functional and usable on desktop, tablet, and mobile phone browsers.
- Back-end (Server-side): A robust framework for handling business logic, user authentication, and data requests.
- Database: A relational database (e.g., PostgreSQL or MySQL) to manage the structured data, including student records, grades, and attendance, ensuring secure storage.
- Deployment Environment: A cloud-based hosting solution (e.g., AWS, Azure, or Google Cloud) to ensure high availability and automatic data backup.
- Version Control: All source code will be managed and maintained on the project's GitHub repository.

20.2. Component Interaction and Layers

The architecture follows a layered approach:

1. Presentation Layer (Client): Handles user input (e.g., a Teacher entering grades) and displays information (e.g., a Student viewing their schedule).
2. Business Logic Layer (Application): Enforces rules, such as checking for scheduling conflicts and performing grade calculations.
3. Data Layer (Database): Responsible for storing and retrieving all system data securely.

21. Data Model Expansion: Grade Calculation & Storage

Further detailing the Grade Record Entity, the system must support the required calculation complexities to achieve automated overall grades.

Attribute	Description	Data Type	Notes/Calculation Logic
GradeID	Unique ID for the grade entry.	Integer	Primary Key.
StudentID	Link to the Student.	Integer	Foreign Key.
SubjectID	Link to the specific subject.	Integer	Foreign Key.
AssignmentName	Name of the assessment (e.g., "Midterm Exam," "Homework 5").	String	
MaxScore	Maximum possible points for the assignment.	Float	
ScoreAchieved	Raw score achieved by the student.	Float	Recorded by Teacher ¹³ .
Category	Type of assessment (e.g., Homework, Test, Participation).	Enum	Used for fixed weighting (e.g., Homework 20%, Tests 50%) ¹⁴ .
CustomWeight	Customizable weighting factor	Float	Used for flexible weighting calculations

Attribute	Description	Data Type	Notes/Calculation Logic
	specific to this individual assignment instance.		(if selected by teacher) ¹⁵ .
SubmissionDate	Date the grade was recorded.	Date	Supports generating grade reports for review ¹⁶ .

22. Final Review and Authorization (Sign-off)

This final section confirms that the documented requirements are formally approved by the key executive stakeholders before development proceeds.

The undersigned stakeholders certify that they have reviewed and approved the requirements outlined in this Software Requirements Specification (SRS) for the School Management System (SMS). These requirements form the baseline for system design, development, and testing activities.

Stakeholder Name	Role/Title	Date of Approval	Signature
School Principal	Executive Stakeholder (Making decisions according to system data) ¹⁷		
School Administrator	Internal Stakeholder (Full access to manage all modules) ¹⁸		

Stakeholder Name	Role/Title	Date of Approval	Signature
Lead Developer	External Stakeholder (Upgrading the system features) ¹⁹		

 Software Requirements Specification (SRS) - School Management System
(Continued)

23. Detailed Fit Criteria from Elicitation Data

This section translates specific preferences gathered from the Questionnaires ¹ into quantifiable, testable criteria (Fit Criteria), ensuring the system meets user expectations for usability and core functionality.

Question/Area	Preference/Choice	Fit Criterion (Measurable Test)	Related FR/NFR
Attendance Input Method ²	List-based check box for all students ³ (fastest for full attendance)	The default attendance recording interface must be a list view with quick checkbox/toggle input for marking 100% of students present/absent for a class.	FR-ATT- 100
Urgent Communication ⁴	Instant Email notification and	The system must successfully send an SMS message to a	FR- COM- 130,

Question/Area	Preference/Choice	Fit Criterion (Measurable Test)	Related FR/NFR
	SMS/Text message 5555	verified number within \$5\$ seconds of an Admin posting an "Urgent" announcement.	FR- COM- 230
Non-Urgent Communication 6	Notification feed on the dashboard 7	Non-urgent notifications should appear in the user's dashboard feed upon login, without triggering a disruptive pop-up or banner.	FR- COM- 120
Search Functionality 8	Advanced filter search (by date, subject, user role) 9	Search functionality within the Announcement and Record modules must allow filtering by at least three criteria (date range, subject/class, and recipient role) simultaneously.	FR- DMR- 110

Question/Area	Preference/Choice	Fit Criterion (Measurable Test)	Related FR/NFR
Grade Viewing Organization ¹⁰	By Course: Grouped by current subject ¹¹	The default display for a Student viewing their grades must first group scores by the corresponding course/subject name.	FR- GRD- 120
Export Format Priority ¹²	Excel (formatted file) and CSV (simple text file) ¹³	The system must offer the option to export generated reports (grades/attendance) into both the Excel format and the CSV format.	FR-RPT- 140

24. Formalizing Team Structure and Roles

To clarify responsibilities, the following roles are designated for the development team ¹⁴:

Name	Student ID	Designated Role	Primary Responsibilities (Example)
Marwan Khaled ¹⁵	23- 101296 ¹⁶	Project Lead / Requirements Analyst	Coordination, Stakeholder Communication, Requirements Validation ¹⁷ .

Name	Student ID	Designated Role	Primary Responsibilities (Example)
Youssef Idrees ¹⁸	23-101077 ¹⁹	Lead Developer / Infrastructure	Core Coding, GitHub Management ²⁰ , System Architecture, Security Implementation.
Hany Yasser ²¹	23-101107 ²²	Backend Developer / Database	Logic Implementation (Scheduling, Grading), Data Modeling, Backup implementation ²³ .
Omar Elsayed ²⁴	23-101091 ²⁵	Frontend Developer / UI/UX	Interface Design (Prototyping), Accessibility (NFR-ACC-100), Client-side logic.
Yassin Sameh ²⁶	23-101192 ²⁷	QA Tester / Documentation	Testing Acceptance Criteria, User Story Writing, Documentation Review.

25. Requirements Elicitation Report (Summary)

The objective of the elicitation process was to gather the necessary information to understand user needs for the School Management System, particularly for new modules like Class Scheduling and Communication²⁸.

- Completeness: Five techniques were deliberately selected (Interviews, Questionnaires, Documents, Observation, Prototyping) to ensure complete and correct requirements from all user types (admin, teachers, students)²⁹.

- Confirmation: The process confirmed that the central problem is the reliance on "paper or old systems" ³⁰ leading to slow communication and organizational challenges³¹.
- Value Proposition: The core value derived from the elicitation is the ability to save time, maintain organization, and facilitate easier communication by storing everything digitally and providing easy access to learning materials³²³²³²³².

This Elicitation Report validates the foundation of the functional and non-functional requirements detailed throughout this SRS.

I will continue the SRS document by adding a section detailing the testable requirements based on the Questionnaire responses (Fit Criteria) and formalizing the team roles and the required Elicitation Report.

Software Requirements Specification (SRS) - School Management System (Continued)

23. Detailed Fit Criteria from Elicitation Data

This section translates specific preferences gathered from the Questionnaires ¹ into quantifiable, testable criteria (Fit Criteria), ensuring the system meets user expectations for usability and core functionality.

Question/Area	Preference/Choice	Fit Criterion (Measurable Test)	Related FR/NFR
Attendance Input Method	List-based check box for all students ³ (fastest for full attendance)	The default attendance recording interface must be a list view with quick checkbox/toggle input for marking 100%	FR-ATT- 100

Question/Area	Preference/Choice	Fit Criterion (Measurable Test)	Related FR/NFR
		of students present/absent for a class.	
Urgent Communication	Instant Email notification and SMS/Text message	The system must successfully send an SMS message to a verified number within \$5\$ seconds of an Admin posting an "Urgent" announcement.	FR- COM- 130, FR- COM- 230
Non-Urgent Communication	Notification feed on the dashboard	Non-urgent notifications should appear in the user's dashboard feed upon login, without triggering a disruptive pop-up or banner.	FR- COM- 120
Search Functionality	Advanced filter search (by date, subject, user role)	Search functionality within the Announcement and Record modules must allow filtering by at least three criteria (date range,	FR- DMR- 110

Question/Area	Preference/Choice	Fit Criterion (Measurable Test)	Related FR/NFR
		subject/class, and recipient role) simultaneously.	
Grade Viewing Organization	By Course: Grouped by current subject ¹¹	The default display for a Student viewing their grades must first group scores by the corresponding course/subject name.	FR-GRD-120
Export Format Priority	Excel (formatted file) and CSV (simple text file)	The system must offer the option to export generated reports (grades/attendance) into both the Excel format and the CSV format.	FR-RPT-140

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- Confirmation: The process confirmed that the central problem is the reliance on "paper or old systems" ³⁰leading to slow communication and organizational challenges³¹.
- Value Proposition: The core value derived from the elicitation is the ability to save time, maintain organization, and facilitate easier communication by storing everything digitally and providing easy access to learning materials³²³²³²³².

27. Elicitation Artifacts: Interview Agenda

The structured agenda used for semi-structured interviews with key stakeholders is reproduced below, illustrating the focused inquiries used to extract specific requirements for the system modules ¹.

27.1. Internal Stakeholders (Users) Interview Questions

Role	Area of Focus	Key Questions and Topics	Related FR/NFR
Teachers	Attendance, Grades, Communication	<i>Attendance Workflow:</i> Describe the current step-by-step process for taking attendance ² . <i>Grade Entry Pain Points:</i> What specific calculations (e.g., weighted averages) do you need the system to handle	FR-ATT-100, NFR-CALC-100, FR-COM-140

Role	Area of Focus	Key Questions and Topics	Related FR/NFR
		automatically³? <i>Communication Needs:</i> What features would help you "communicate better" within the system⁴?	
Students	Grade Viewing, Schedule, Announcements	<i>Grade Viewing Experience:</i> What specific information are you looking for (e.g., final course average, missing work)⁵? <i>Announcement Consumption:</i> What's the best way for the system to deliver announcements to you (in-system notification or email)⁶? <i>System Access:</i> How fast must the page load before you become frustrated⁷?	FR-GRD-120, FR-COM-130, Sec 3.2

27.2. External Stakeholders (Client/School) Interview Questions

Role	Area of Focus	Key Questions and Topics	Related FR/NFR
Admin/Principal/Owner	Management, Reporting, Security	<i>Data Management Goals:</i> What are the top strategic benefits the school expects to gain (e.g., better parent engagement, increased efficiency)⁸? <i>Report Customization:</i> Do you need to see trends, comparative data (e.g., class vs. class), or just simple individual summaries⁹? <i>User Onboarding/Offboarding:</i> Describe the process for creating, updating, and disabling user accounts (Admin, Teacher, Student)¹⁰.	FR-RPT-100, FR-UAM-100, FR-UAM-120

28. Elicitation Artifacts: Questionnaire Content

The content of the questionnaire, distributed to teachers and students to gather scale and preference data, is summarized below .

Section	Key Question	Response Options / Insights	Related FR/NFR
General and Access	Where would you primarily access the system from?	Desktop/Laptop, Tablet/iPad, Mobile Phone Browser ¹² .	NFR-ACC-100 (Cross-Device Support)
Teacher-Specific	How would you prefer to see the scores weighted?	Pre-defined categories (e.g., Homework 20%), Customizable weights, Total points system .	NFR-CALC-100 (Flexible Weighting)
Student-Specific	When checking schedules, which format is most usable?	A simple daily/weekly list view, An interactive calendar view , A graphical timeline.	NFR-ACC-110 (Schedule Visualization)
Communication	When receiving urgent school announcements, do you prefer...?	Instant Email notification, In-system banner/pop-up alert, SMS/Text message ¹⁵ .	FR-COM-130 (Multi-channel Urgent Delivery)

Section	Key Question	Response Options / Insights	Related FR/NFR
Efficiency	If the internet connection is temporarily lost while entering data, which behavior is most critical?	System saves data locally and synchronizes once reconnected ¹⁶ .	NFR-DR-110 (Data Loss Prevention)

29. Test Plan Overview

This section outlines the strategy for testing the School Management System (SMS) to ensure it meets all functional and non-functional requirements detailed in this SRS.

29.1. Testing Objectives

The primary goal of testing is to verify that the system is:

1. Functionally Complete: All defined Use Cases (e.g., Attendance Recording, Scheduling Conflict Check) perform as specified in the detailed flows.
2. Compliant with Non-Functional Requirements (NFRs): The system meets expectations for speed, security, and usability.
3. Secure: Role-based access control (Authorization) is strictly enforced to restrict unauthorized data access.

29.2. Key Test Categories

Test Category	Focus Area	Requirement Source	Test Tool/Method
Functional Testing	Data Management (CRUD), Workflow Accuracy (Scheduling logic), Communication Flows (Multi-channel delivery) ⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴ .	FR-UAM-100, FR-SCH-200, FR-COM-200	Manual Testing, Scenario Execution (Given-When-Then criteria) ⁵ .
Security Testing	Authentication, Authorization (Role-based access), Auditing ⁶⁶⁶ .	NFR-SEC-200 (Restrict Unauthorized Access), NFR-SEC-210 (Activity Log Detail) ⁷⁷⁷⁷ .	Penetration testing (basic), Role matrix verification (ensure a Student cannot access Admin tools) ⁸ .
Performance Testing	Transaction speed (Grade updating), Page loading speed ⁹ .	NFR-CALC-110 (Instant Final Average), Sec 3.2 (Page Load Speed) ¹⁰¹⁰¹⁰ .	Automated Load Testing, Stopwatch measurement (Fit Criteria verification) ¹¹ .
Usability (UX) Testing	Cross-device responsiveness, Interface clarity	NFR-ACC-100 (Cross-Device Support), Sec	Browser testing across minimum screen widths,

Test Category	Focus Area	Requirement Source	Test Tool/Method
	(labels, guidance) ¹²¹²¹² .	3.2 (Look and Feel) ¹³¹³¹³ .	User acceptance testing (UAT) with mockups ¹⁴ .

29.3. Traceability to Acceptance Criteria

Every User Story and Functional Requirement must be traceable to one or more Test Cases, as illustrated in the Requirements Traceability Matrix (RTM)¹⁵. This guarantees that all required functionality will be tested and validated prior to deployment.

I will continue the SRS document by formalizing the structure around the requirements documentation itself, aligning it with high-quality software engineering standards. This includes adding a document history and version control section.

 Software Requirements Specification (SRS) - School Management System (Continued)

30. Document History and Version Control

This section ensures proper configuration management for the requirements documentation, adhering to professional standards for accountability and traceability.

Version	Date	Author	Description of Changes
0.1	2025-12-11	Project Team	Initial Draft: Document structure established, core goals and scope defined, initial functional and non-functional requirements populated from Deliverable 2 (User Stories/Use Cases) and Technical Report.
0.2	2025-12-11	Project Team	Expansion: Detailed flows for Class Scheduling and Communication added; preliminary acceptance criteria and data model included.
0.3	2025-12-11	Project Team	Finalization: Integration of Elicitation data (Interviews, Questionnaires), formalization of team roles, Test Plan Overview, and Document History added.

31. Requirements Prioritization

Based on the Mandates ¹, Stakeholder Needs ², and Elicitation results ³³³³³³³³³, the requirements are prioritized to guide the iterative development process.

31.1. Core Prioritization Categories (High, Medium, Low)

Priority Level	Definition	Example Requirements
High (Mandatory)	Core functions absolutely necessary for the system to fulfill its main purpose ("Must-haves"). Failure to	FR-ATT-100 (Record Attendance) ⁴⁴⁴⁴ , FR-GRD-100 (Enter Grades) ⁵ , FR-UAM-100 (Manage User Accounts) ⁶ ,

Priority Level	Definition	Example Requirements
	implement stops the project.	NFR-SEC-200 (Restrict Unauthorized Access) ⁷ .
Medium (Desirable)	Important features that provide significant value (Value proposition) and competitive advantage, but the system can function without them in a minimal state.	FR-SCH-120 (Check for Conflicts), FR-RPT-130 (Advanced Search/Filter), FR-COM-140 (Built-in Messaging).
Low (Optional)	Features or design elements that would enhance user experience or compliance but can be deferred until later phases or iterations.	FR-UAM-130 (Password Reset Functionality) ⁸ , NFR-ACC-110 (Interactive Calendar View).

31.2. Prioritized Requirement Examples

Requirement ID	Description	Priority	Rationale (Based on Source)
FR-GRD-110	System shall calculate and display overall grades/averages automatically.	High	Essential to achieve the main goal of keeping data organized and easy to find ⁹⁹⁹⁹ .

Requirement ID	Description	Priority	Rationale (Based on Source)
NFR-SYS-110	System shall require an active internet connection.	High	A defined limitation and constraint for the current scope ¹⁰ .
NFR-DR-100	System shall automatically back up data periodically.	High	Critical for System Management and Security ¹¹ .
FR-MAT-100	Teachers shall upload study materials.	Medium	Provides students with access to study materials, fulfilling a main goal, but not critical to the core data management loop ¹² .
